# Team Details

**Team Name:** GrayScale

| SR. NO | ROLE | NAME | ACADEMIC YEAR |
|---|---|---|---|
| 1 | **Team Leader** | *Nisheet Lad* | *3rd Year* |
| 2 | Member 1 | *Deven Mahajan* | *3rd Year* |
| 3 | Member 2 | *Osh Manoj Kumar* | *3rd Year* |
| 4 | Member 3 | *Satyam Katkar* | *3rd Year* |

***A team can have up to 4 members including the team leader. Add rows if necessary.***

**COLLEGE NAME**

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**TEAM LEADER CONTACT NUMBER**

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# Problem Statement Addressed

## KLA PS01 — AI-Based Restoration of Degraded Images

DESCRIPTION / DETAILS

### WHY THIS MATTERS IN SEMICONDUCTOR FABS

- Microscopic inspection depends on fine structural detail.
- Noise removal alone is insufficient if restoration introduces blur or artificial texture.
- The model must generalize to unseen image sources, making OOD robustness critical.

### Semiconductor inspection images can suffer from

- **Speckle noise** — signal-dependent multiplicative noise that can push pixel values outside the valid range.
- **Additive Gaussian noise** — further obscures edges and fine structures.
- **2× spatial downsampling** — removes high-frequency detail required for accurate inspection

### 1 · SPECKLE NOISE

- Multiplicative, signal-dependent grain
- Pushes values beyond the true range
- Removed without blurring — blur destroys the detail we are paid to recover

Our observation: measured speckle variation (~0.24) dominates the additive noise range (0.001–0.04), so restoration must prioritize despeckling without destroying detail.

### 2 · 2× DOWNSAMPLING

— 256×256 → 128×128 in this release
— 512×512 → 256×256 in KLA's deck
— Lost high-frequency detail reconstructed at exactly 2× the input

# Idea Description

A degradation-aware AI restoration pipeline that learns to remove dominant speckle noise while reconstructing lost spatial detail.

KEY CONCEPT & APPROACH

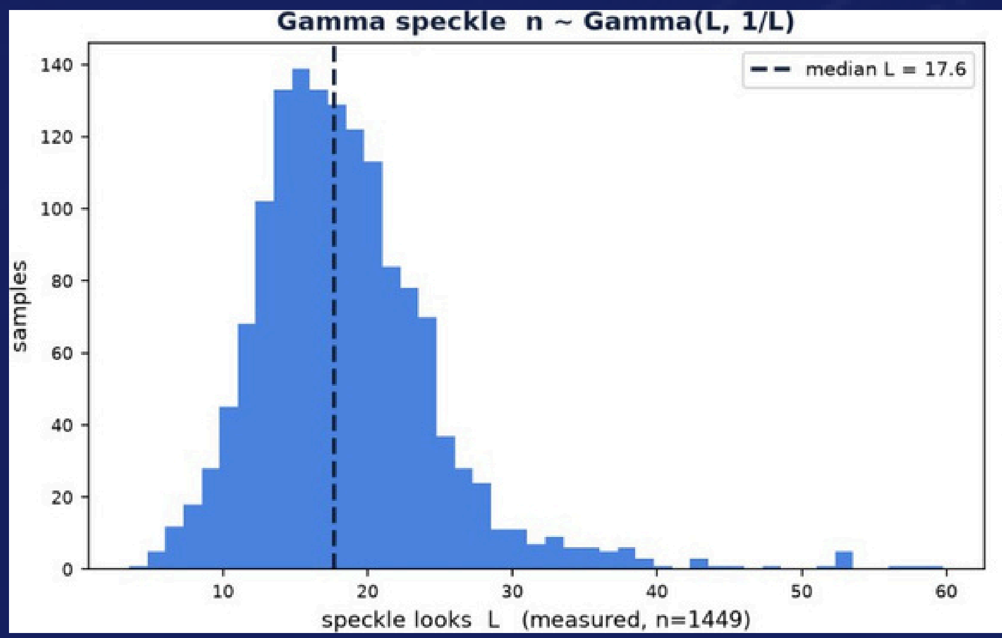
*A degradation-aware AI restoration pipeline that learns to remove dominant speckle noise while reconstructing lost spatial detail.*

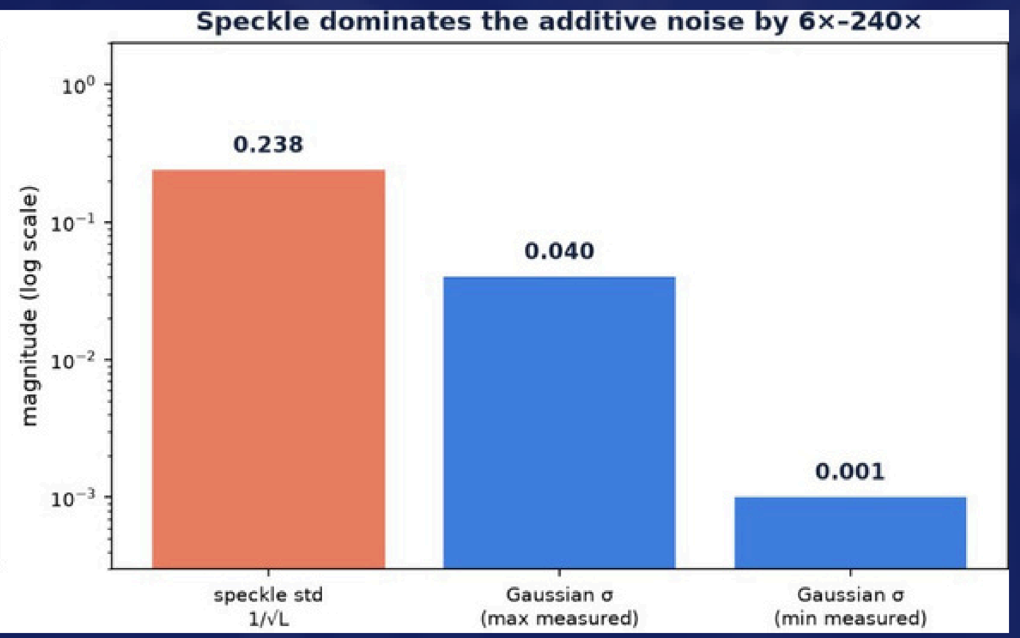
- *Reverse-engineer the degradation process from paired data.*
- *Model multiplicative Gamma speckle + additive Gaussian noise + cubic downsampling.*
- *Train a restoration network directly on realistically degraded images.*
- *Preserve fine structures using perceptual, structural and frequency-domain constraint*

SOLUTION OVERVIEW

**MODEL CHOICE — NAFNet-w32**

- No activation functions; LayerNorm generalises well out-of-distribution
- Best quality-per-FLOP among general restoration backbones
- Fully convolutional → both input sizes work unchanged
- SAFMN (~0.24 M) kept as the fast point on the Pareto curve



# Proposed Solution

## Architecture · training strategy · loss design · augmentation

**INFERENCE PATH**

**Degraded input** .npy float32 → **Reflect-pad** to mult. of 16 → **NAFNet-w32** at LR resolution → **PixelShuffle ×2** ICNR init → **+ bicubic anchor** residual → **Output** exactly 2×

**TRAINING DATA FLOW**

**Clean HR source** KLA GT + external → **× Gamma(L,1/L)** speckle at HR → **+ N(0, σ²)** additive → **Cubic ×2 decimation** LAST — measured → **Paired (LR, HR)** float32

### TRAINING STRATEGY
- AdamW, lr 2e-4, cosine + warmup
- bf16 autocast (no GradScaler)
- EMA decay 0.999, evaluated with EMA
- Split by source_id = index // 4

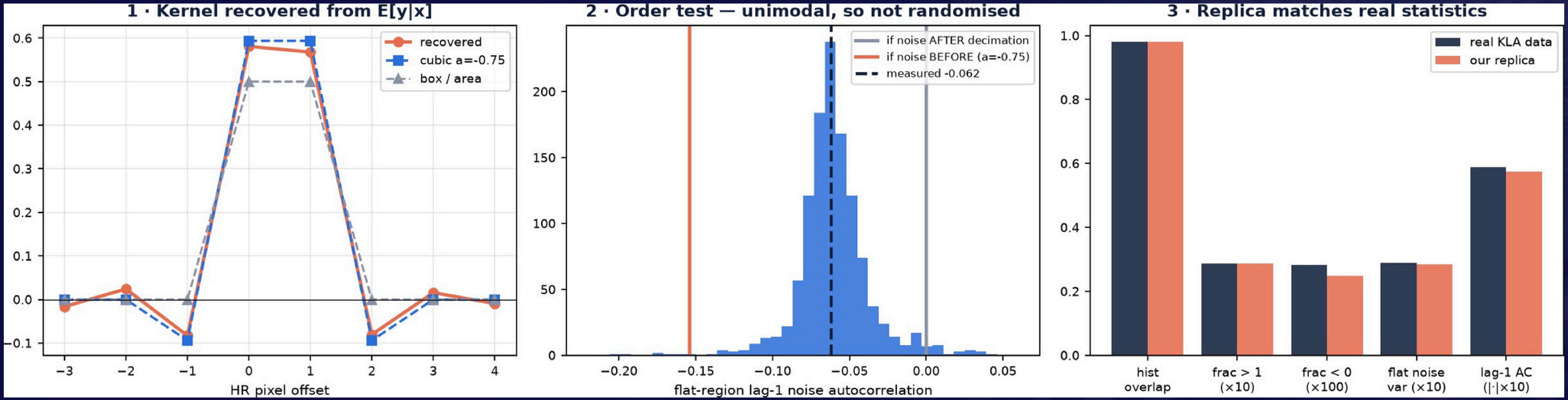
### COMPOSITE LOSS
- 1.0 · Charbonnier
- 0.2 · (1 − MS-SSIM)
- 0.1 · FFT (L1 in frequency)
- 0.05 · gradient + 0.01 · VGG

### AUGMENTATION
- 8-fold dihedral on paired crops
- L ~ 8–40, σ ~ log-uniform(1e-3, 4e-2
- Kernel a ~ U(−0.75, −0.45)
- 50/50 KLA pairs vs external content

**Never clipped. Processing stays at LR resolution throughout — upsampling first would cost ~4× the compute for no quality gain.**

# Innovation and Uniqueness

## The degradation was reverse-engineered from the data, not assumed




### 1 · RECONSTRUCTED DEGRADATION

- No metadata ships with the dataset — we recovered the model from the pairs
- Kernel regressed directly from E[y|x]; box, area, Lanczos and strided decimation all excluded
- Replica hits 0.982 histogram overlap with the real data

### 2 · LOG-DOMAIN UNROLLED NET

- Built from KLA's own appendix reference (Monga et al. 2021) — a slide never presented
- log() turns multiplicative speckle additive, so HQS/ADMM machinery applies
- K analytic data-fidelity steps alternating with a learnt CNN prior

### 3 · SYNTHETIC OOD VALIDATION

- KLA withheld test ground truth — we manufacture it
- Validated replica applied to external corpora gives labelled OOD families
- Degradation-consistency check scores unlabelled test images with no GT

# Impact and Benefits

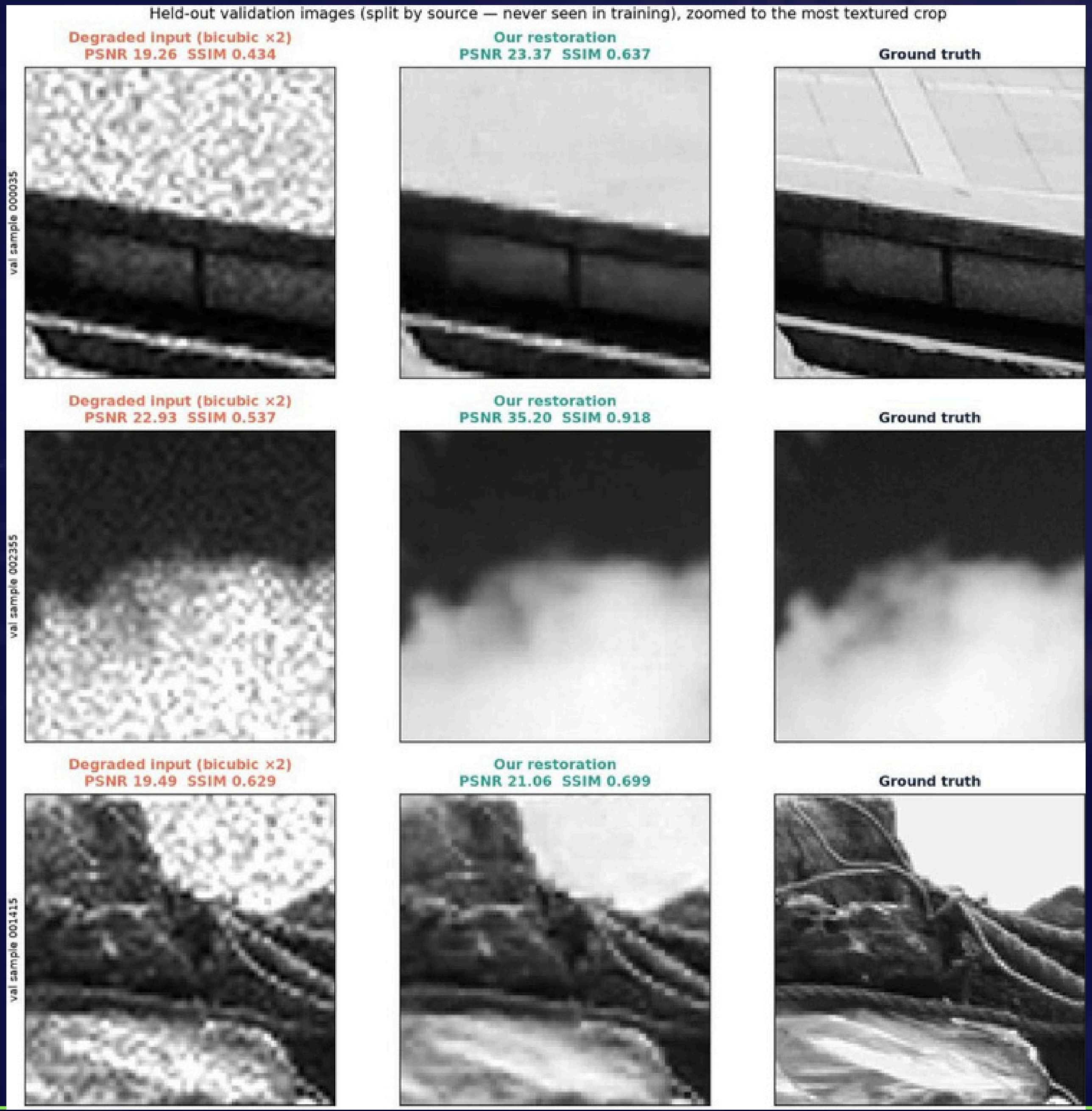
Recover usable inspection detail from noisy, spatially undersampled SEM images without requiring slower acquisition.

### Primary Impact

- Better visibility of fine semiconductor structures.
- Reduced impact of speckle and additive noise.
- Restoration at exactly 2× resolution.
- Computational restoration can complement high-throughput acquisition.
- Better robustness to previously unseen image sources.

### QUANTIFIABLE OUTCOMES

| Metric | Bicubic ×2 | Our NAFNet |
|---|---|---|
| PSNR | 23.067 | **26.415** |
| SSIM | 0.5129 | **0.7333** |
| LPIPS ↓ | 0.4425 | **0.2455** |



# Technology & Feasibility/Methodology Used

Describe the technologies, methodologies, or tools you plan to use to implement your idea.

IMPLEMENTATION STRATEGY

| Component | Choice |
|---|---|
| Framework | PyTorch 2.11.0 + CUDA 12.8 |
| Training GPU | consumer GPU, 8 GB (recent architecture) |
| Benchmark target | NVIDIA H100 (KLA-side) |
| Precision | bf16 autocast (train) · fp16 (inference) |
| Environment | conda, Python 3.11.15, 115 pinned packages |
| Metrics | torchmetrics · lpips · pyiqa · bm3d |

| **Measure** | **Value** |
|---|---|
| Model parameters | 15.24 M |
| Weight file (fp16) | 30.6 MB (8× smaller than ckpt) |
| Training time | 40 epochs, ~1.6 h on an 8 GB GPU |
| Peak training VRAM | 7.1 GB of 8 GB |
| Inference, per image | 22.5 ms end-to-end |
| Full test set (400 img) | 9.0 s · 44.4 img/s |

Software Architecture

Hardware Components

Development Tools

# GitHub & Video Link

### GitHub Repository

https://github.com/Deven1305/GrayScale

### Prototype / Simulation Video

https://drive.google.com/drive/folders/1DkD9Aaw24-iuYF3FvX3nXvvYnU8-5WKj?usp=sharing

# Research and References

## Research Background & Methodology

Briefly describe the research foundation or scientific principles supporting your idea.


1. T. Kumar, R. Brennan, A. Mileo, M. Bendechache. “Image Data Augmentation Approaches: A Comprehensive Survey and Future Directions.” IEEE Access, vol. 12, 2024.
2. L. Zhai, Y. Wang, S. Cui, Y. Zhou. “A Comprehensive Review of Deep Learning-Based Real-World Image Restoration.” IEEE Access, vol. 11, pp. 21049–21067, 2023.
3. J. Terven, D. M. Cordova-Esparza, J. A. Romero-González et al. “A Comprehensive Survey of Loss Functions and Metrics in Deep Learning.” Artificial Intelligence Review 58, 195, 2025.
4. V. Monga, Y. Li, Y. C. Eldar. “Algorithm Unrolling: Interpretable, Efficient Deep Learning for Signal and Image Processing.” IEEE Signal Processing Magazine, vol. 38, no. 2, pp. 18–44, 2021.


## References & Citations

List key papers, articles, or data sources.

**METHODS**


1. L. Chen, X. Chu, X. Zhang, J. Sun. “Simple Baselines for Image Restoration.” ECCV 2022. (NAFNet)
2. K. Zhang, J. Liang, L. Van Gool, R. Timofte. “Designing a Practical Degradation Model for Deep Blind Image Super-Resolution.” ICCV 2021. (BSRGAN)
3. K. Zhang, L. Van Gool, R. Timofte. “Deep Unfolding Network for Image Super-Resolution.” CVPR 2020. (USRNet)
4. L. Sun, J. Dong, J. Tang, J. Pan. “Spatially-Adaptive Feature Modulation for Efficient Image Super-Resolution.” ICCV 2023. (SAFMN)


**DATASETS & TOOLS**


1. E. Agustsson, R. Timofte. “NTIRE 2017 Challenge on Single Image Super-Resolution: Dataset and Study.” CVPRW 2017. (DIV2K)
2. J.-B. Huang, A. Singh, N. Ahuja. “Single Image Super-Resolution from Transformed Self-Exemplars.” CVPR 2015. (Urban100)
3. M. Cimpoi et al. “Describing Textures in the Wild.” CVPR 2014. (DTD)
4. K. Dabov et al. “Image Denoising by Sparse 3-D Transform-Domain Collaborative Filtering.” IEEE TIP, 2007. (BM3D)
5. R. Zhang et al. “The Unreasonable Effectiveness of Deep Features as a Perceptual Metric.” CVPR 2018. (LPIPS)